

Capacitance to Digital Converter using MASH Sigma-Delta ADC Architecture

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Abstract—In the field of MEMS, there is a wide use of capacitive sensors for several applications like pressure sensor, accelerometers etc. Due to this we also need analog to digital converters which are more compact and can provide better gain. Hence, in this work we are designing a Capacitance to Digital converter using MASH Sigma-Delta Architecture in the ADC part which can provide better SNR than the existing methods. MASH Sigma-Delta Architecture provides excellent Noise-Shaping which helps in increasing the circuit efficiency. A switched capacitor circuit is used for capacitance-to-voltage conversion, and the voltage output is fed to the MASH Sigma Delta ADC. Switched capacitance circuits provide better frequency response, and the ADC part provides better noise rejection and gain. In this implementation, we achieved an SNR of 77.62 dB with a capacitive resolution of 78.13 aF. This design is implemented in the Cadence Virtuoso environment using 90nm technology, and the simulation results and performance metrics validate the proposed concept.

Index Terms—Micro-Electro-Mechanical Systems (MEMS), Multi-Stage Noise-Shaping (MASH), Capacitance to digital converter(CDC)

I. INTRODUCTION

With the increase in the use of MEMS, there is a requirement to design efficient signal conditioning circuits in terms of power and gain, and the circuit should provide less noise. Micro-electro-mechanical systems (MEMS) are small, integrated systems or devices that integrate mechanical and electrical elements on a microscale. MEMS devices can have electronics, sensors, actuators, and structures on a single chip. Capacitive sensing is one of the most popular sensing techniques in MEMS. MEMS capacitive sensors function by detecting variations in capacitance brought on by the motion of mechanical components. Because of their high sensitivity, low power consumption, and compatibility with conventional CMOS fabrication processes, capacitive sensors are especially well-liked in MEMS. There are many uses for these sensors. MEMS capacitive accelerometers and gyroscopes are used in consumer electronics to detect tilt, motion, and orientation in wearable technology, smartphones, and gaming consoles.

They are essential to systems like tire pressure monitoring, electronic stability control, and airbag deployment in the automotive sector. MEMS capacitive pressure sensors are also helpful for medical devices because they aid in monitoring parameters like blood pressure and respiratory function. Due to their adaptability and compact size, capacitive sensors in MEMS have become crucial parts of contemporary electronic systems in various industries. This requirement encourages the development of research on signal conditioning circuits that are suitable for MEMS. In this work, we are designing a capacitance-to-digital converter using MASH Sigma-delta ADC. A capacitance-to-digital converter is a circuit that converts the capacitance value into a readable digital value. There are different techniques for converting capacitance to digital conversion. The capacitance is first converted into time and then converted into digital format. The other method is capacitance, which is converted into voltage and digital format. In these methods, time-based methods require a long time for the capacitive measurement. However, they offer a wide range of capacitance measurements with the drawback of low power efficiency and high time consumption. The second type of circuit is capacitance to voltage and voltage to digital conversion. We can use different types of ADCs, such as PM-based, SAR(Successive approximation Register), and Sigma-Delta ADC. [?] [15] [9] [14] [3] PM-based comes under capacitance to time conversion type [13] [8]. SAR-based CDC is susceptible to high noise even after many noise reduction techniques. Coming to Sigma-Delta ADCs, [4] they provide high resolution with less noise, making it efficient in all the available methods. [9] [11] [6] [7] [10] [2] [12] To even get a better noise shaping and to overcome the drawbacks of Sigma-Delta ADC a novel architecture came into existence which is MASH Sigma-Delta ADC. This is the latest Sigma-Delta ADC architecture, a cascaded structure of multiple sigma-delta modulators to improve noise shaping and resolution. MASH ADCs are commonly used in high-speed, high-resolution communication systems and sensor-interfacing applications.

MASH architecture consists of numerous cascaded sigma-delta modulators, where each stage processes the quantization error from the previous stage to improve overall noise shaping. The output of each stage is digitally processed to cancel quantization noise, allowing higher-order noise shaping without the stability issues of a single high-order loop. MASH achieves high resolution, better SNR, and increased linearity by splitting noise shaping across multiple stages, making it ideal for precision ADC applications. A sophisticated type of sigma-delta modulation, the MASH (Multi-stage noise Shaping) Sigma-Delta architecture enhances performance by integrating several low-order sigma-delta modulators in a cascaded structure. In a MASH architecture, each stage applies noise shaping to the quantization error of the stage before it. The main idea is that a single high-order sigma-delta modulator may have stability problems even though it can achieve excellent noise shaping. The MASH technique gets around this by carefully combining several stable low-order stages, producing high-order noise shaping without running the risk of instability.

II. PROPOSED ARCHITECTURE

This study combined a capacitance-to-digital converter with a 3-2 MASH Sigma-delta architecture for analog-to-digital conversion. The MASH Sigma-delta ADC receives the output of a switched capacitor circuit that first transforms the capacitance into voltage. The block diagram of MASH Sigma-delta ADC is shown in Fig. 1.

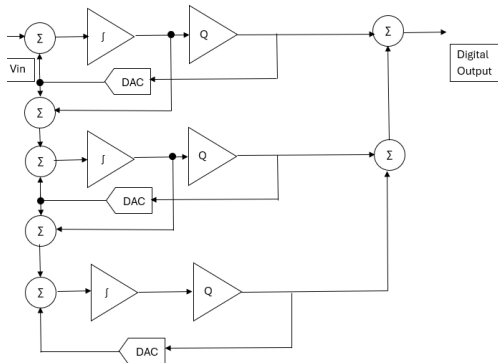


Fig. 1: MASH Sigma-Delta Architecture

This circuit is a multi-stage delta-sigma analog-to-digital converter (ADC), more precisely a cascade architecture that employs noise shaping to attain high accuracy and resolution. Its three main stages are a digital-to-analog converter (DAC), a quantizer, an integrator, and a differencing stage. The output is produced by digitally combining the outputs of each stage in order to enhance signal-to-noise performance.

a) First Stage:: The input signal V_{in} is fed into the first summing block (Σ) after being subtracted by the feedback signal from the DAC. After the resultant error signal has been integrated by the integrator (I), it is quantized by the quantizer, which is typically a 1-bit or multi-bit ADC. The digital output is looped back through a DAC to create the feedback signal.

The quantization noise will be forced to higher frequencies by this step.

b) Second Stage:: The quantization noise generated by the first stage is removed by computing the difference between the input and DAC output of the first quantizer, also referred to as the quantization residue. The second summing block (Σ) integrates, quantizes, and processes this residue signal similarly before feeding it back using a different DAC.

c) Third Stage:: The quantization noise from the second stage is extracted and processed using a similar process. It is obtained by the third integrator-quantizer-DAC chain. By capturing any residual noise that the previous stage missed, this step further enhances the system's overall performance and resolution. The digital outputs from each step are combined to create the final high-resolution digital output. The signal-to-noise ratio (SNR) is significantly improved by the cumulative noise shaping effect.

III. IMPLEMENTATION IN CAPACITANCE TO DIGITAL CONVERSION

1) Design of Capacitance to voltage converter: In this work, the capacitance to voltage converter is implemented using a Switched capacitor circuit [1], and the output of this is fed to the MASH Sigma-Delta Converter as shown in Fig. 2.

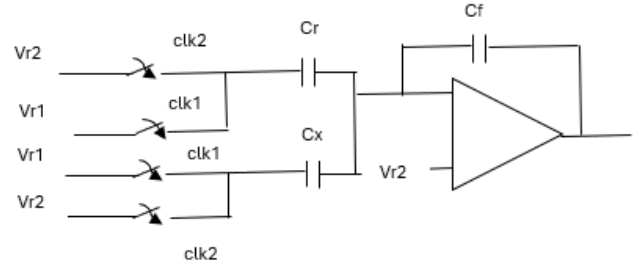


Fig. 2: Switched Capacitor circuit

The CVC uses an operational amplifier (Op-amp) in conjunction with a capacitive network that consists of sensor capacitors (C_x), designed capacitors (C_r and C_f), and more. The Op amp circuit is connected to the input sensor capacitor (C_x), which senses changes in capacitance brought on by changing physical parameters (such as pressure or humidity).

Mechanism of Operation: By charging or discharging the capacitors in response to a clock signal ($ck1$ or $ck2$), the Op-amp configuration modifies the output voltage (V_o) depending on the capacitance measured by C_x . In particular, when $ck2$ is high, C_f stores the Op amp's offset voltage, and C_x is charged using the reference voltage ($Vr2$). On the other hand, C_f connects to the output when $ck1$ is high, affecting the output voltage according to charge conservation principles.

Derivation of Output Voltage: The relationship between the sensor's capacitance and the reference capacitors yields the

output voltage, V_o . The formula that defines the voltage output is

$$V_o = (Cx - Cr/Cf) * Vr$$

This formula illustrates how variations in capacitance at C_x concerning the intended capacitors affect the output voltage.

2) *Designing of MASH Sigma-Delta Blocks:* The basic blocks of MASH 3-2 Sigma-Delta are an integrator, DAC, and a Quantizer.

a) *Design of Quantizer:* Fig 3 shows the quantizer circuit, which consists of four operational amplifiers set up as comparators used in the quantizer circuit. A resistor ladder network between two voltage sources sets the reference voltage, which is then compared to a standard analog input voltage by each opamp. At their nodes, these resistors produce reference voltages that are uniformly spaced.

All of the opamps non-inverting inputs receive the analog input, and the inverting inputs are linked to the corresponding resistor ladder nodes. The corresponding opamp output switches high when the input voltage is higher than a comparator's reference voltage and low when it is lower. The result is a collection of digital outputs reflecting the analog input's quantized level.

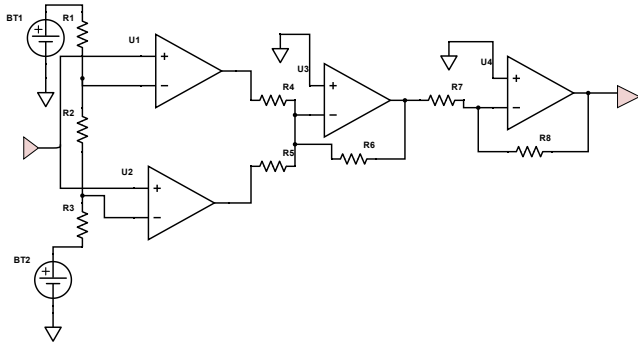


Fig. 3: Quantizer

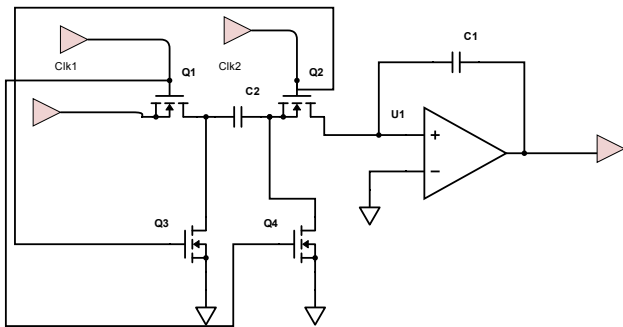


Fig. 4: Clocked parasitic-insensitive Integrator

b) *Design of Integrator:* The parasitic insensitive Integrator's shown in Fig. 4 depends on switched capacitor (SC) circuits [5] , which successfully substitute capacitors and

switches for conventional resistors. Operational amplifiers (op-amps) in a two-stage topology are incorporated into this design to improve overall performance through increased bandwidth and gain. Its fundamental working principle is the switched capacitor principle, which states that clock signals determine when capacitors should be charged and discharged. These switches enable the clock frequency and selected capacitance values to determine the circuit's effective resistance by regulating the charge flow to and from the capacitors. During operation, a capacitor connected to the op-amp's positive terminal is charged as clock signals trigger switches. The capacitor discharges through the circuit when another switch is triggered by a different clock signal, which causes the input signal to integrate gradually. This process results in a voltage output that reflects the integral value. Additionally, the Integrator's output voltage frequently feeds back into the op-amp's input, forming a feedback loop that improves stability and accuracy. Using this feedback mechanism, the circuit can adapt to the effects of parasitic capacitances and continue to function consistently even when component values or operating conditions change.

c) *Design of DAC:* Fig. represents a single-bit current-steering DAC [16]. The operation is carried out as a constant current source is switched between two output paths by a single-bit Current-Steering DAC converter in response to a single-bit digital input (0 or 1). A steering switch, a current source, and two output nodes—usually called output and ground—are the primary parts of this DAC. The switch directs the reference current toward the positive output, producing a corresponding current when the digital input is set to '1'. The current is directed to the negative output (typically connected to the ground) when the input is set to '0'. The digital input chooses the current path because the source stays constant. Usually, the output voltage is generated across the load that is connected to the output node.

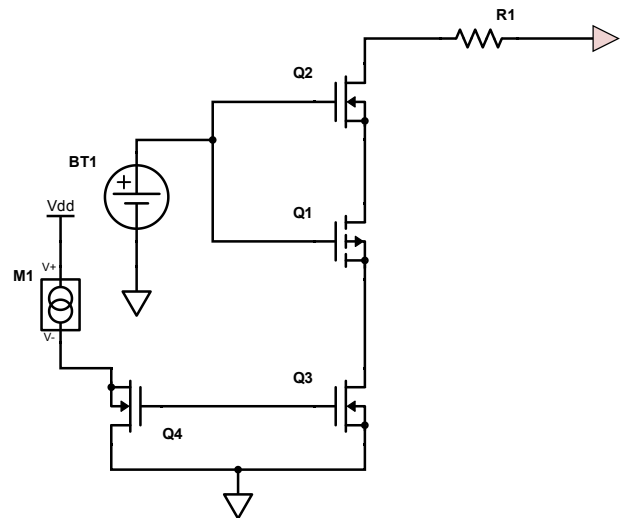


Fig. 5: 1-bit Current-steering DAC

d) *Overall CDC*: The output voltage of the capacitance-to-voltage converter feeds straight into the MASH 3-2 ADC's first modulator stage and represents the instantaneous capacitance value as depicted in Fig. 6. This architecture's first third-order sigma-delta modulator creates quantization noise and digitizes the input voltage with high noise shaping. This noise is transferred to a secondary 2nd-order modulator, which reshapes and digitizes the quantization error rather than letting it deteriorate performance. Due to this cascading and digital error correction technique, the system can maintain high resolution and stability without the complexity of a high-order single-loop modulator. The satisfactory resolution and noise suppression of the MASH ADC combines with the accuracy of the C-to-V stage to create a robust and high-performing CDC, making the integration extremely effective.



Fig. 6: Capacitance to digital conversion

IV. MEASUREMENT RESULTS

This circuit is implemented in Cadence Virtuoso in 90nm and the simulation results are taken from the Capacitance to Digital converter output as depicted in Fig. 7.

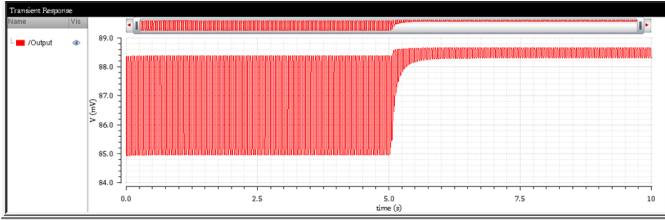


Fig. 7: Simulation results of CDC

The V_{ref} applied for the circuit is 3V and the reference capacitor C_{ref} is set as 10 fF. The capacitance to voltage converter circuit can measure the capacitance from 20 fF to 1700 fF.

$$C_{res} = \frac{C_{max} - C_{min}}{2^N}$$

$$SNR = 20 \cdot \log_{10} \left(\frac{C_{max} - C_{min}}{2\sqrt{2} \cdot C_{res}} \right)$$

$$ENOB = \frac{SNR - 1.76}{6.02}$$

From the above values we computed the parameters of the circuit as presented in Table. I.

Supply voltage	3V
Capacitance Resolution	78.13 aF
Signal to Noise Ratio	77.62 dB
ENOB	12.6 bits

TABLE I: Tabulation of results

Fig. 8 shows the Power Spectral Density (PSD) plot of the Capacitance to Digital Converter.

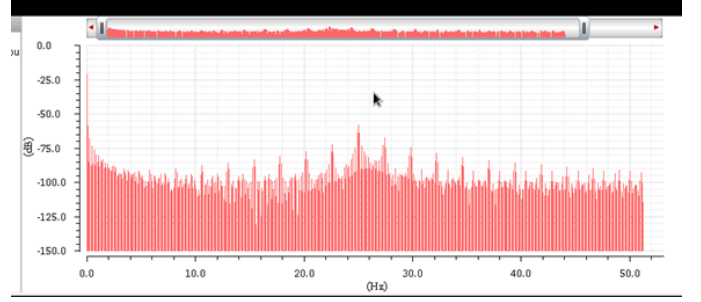


Fig. 8: Power Spectral Density Plot of CDC

V. CONCLUSION

In conclusion, the Capacitance-to-Digital Converter (CDC) measurement results using a 3-2 MASH sigma-delta architecture show remarkable noise immunity and precision performance. The system converts minor capacitance variations into digital values with high accuracy, resulting in minimal distortion and high fidelity in the output, thanks to its Signal-to-Noise Ratio (SNR) of 77.62 dB. Due to its ability to shape noise, the 3-2 MASH architecture substantially improves the dynamic range and lowers quantization noise. These findings demonstrate that the CDC can accurately and consistently measure capacitance with high sensitivity and resolution, which makes it a good choice for applications requiring accurate and consistent readings. A robust architecture and high SNR together also indicate the possibility of outstanding performance in the presence of noise or interference.

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